

DC-“Born Again”: En/De-Roling, Character Identification, & VR Effects

Shane L Burrell Jr*

University of Oregon

ABSTRACT

With the rise of the metaverse, humans are increasingly spending time in mediated spaces, emphasizing the importance to understand how immersive experiences shape individual and collective behavior in various contexts (e.g., enterprise, entertainment). Two particularly important factors emerging in metaverse technologies relate to how users of virtual reality (VR) adapt and relinquish their character roles (en/de-roling), and the extent to which they identify with said roles. Despite the prominent reliance of character roles in most VR experiences, and its noted influence on aspects of user experience (UX), little is understood about (a) how users adapt and relinquish their roles, and (b) how this process may shape affective and cognitive responses through altered identification. To address this gap, the current work employs a mixed-methods approach to understand how VR effects are influenced by these two factors.

Keywords: Virtual reality, identification, embodiment, presence, affect theory, en/de-roling.

1 INTRODUCTION

Virtual reality (VR) is known as a facet of extended reality (XR) technologies (Pimentel et al., 2022), and with its growing popularity in education, commerce, practice, and most important, enthusiast, understanding how VR and its effects manifest through character identification becomes imperative. My research seeks to understand the relationship VR effects have on individuals who adapt and relinquish their character roles in virtual environments (VE). Unlike scholarship on traditional media, research on the psychological mechanisms through which XR content influences attitudes, behaviors, and cognitions remain ambiguous (Gualeni et al., 2021). Understanding how VR affects users becomes imperative as immersive technologies grow and become more integrated in the daily lives of those developing personas in VEs (Nowak, et al., 2018).

2 LITERATURE REVIEW

2.1 Affect Theory and VR

Affect theory provides context in the immersive experiences users of VR have when participating in VR simulations. Tomkins (1962) gave the initial understanding of affect by defining it as a “the biological utility of affects with sharply peaked profiles, such as startle, fear and excitement” (p. 276). Initially, the affects Tomkins (1962) identified were all biological functions, although with more research in this theory, affect is known for its understanding of how emotions come from media and how those

emotions influence attitude and behavior changes (Schaefer, 2019). Affect theory allows for individual experiences of VR users to be explained not only through their biological receptors of feelings but by their emotional responses of the VR experience and how those experiences has shaped their attitudes and behaviors going forward.

2.2 Identification and VR

A review of VR and the concept of identification is needed to situate my research on VR and VR effects. Identification—how stories transport users into a story, character role, and alternative world (Cohen et al., 2016; Nowak et al., 2018)—plays a large role in how VR can immerse its users. VR environments expose users to highly immersive simulations, where they can directly interact with characters. Identification is how media viewers receive emotions from a simulation (Cohen et al., 2016); however, VR users directly interact or play as the characters in the simulation. Therefore, maybe identification occurs in VR as VR users interact as their character. Although this is speculative and VR users may not inhabit their character as their character was designed, they often have a “...deep appreciation for the character’s feelings, goals, and understanding” (Cohen, et al., 2016, p. 184). This understanding of identification not only explains how users relate to their characters in VR, but it also raises questions about VR effects.

Character identification, although like identification, presents a different part of identification, much like avatar identification. Character identification specifically represents and understanding how the users receives and negotiates emotions from a virtual simulation, where an individual is participating in the actions of the simulation and not just simply viewing a screen (Cohen et al., 2016; Nowak et al., 2018). This definition of character identification gives some insight to processes of en-roling and de-roling and how these processes are influences by identification, embodiment, and presence.

2.3 Presence & Embodiment in VR

Embodiment is a form of “body transfer,” or, more precisely, “the ensemble of sensations that arise in conjunction with being inside, having, and controlling a body especially in relation to virtual reality application,” (Kiltner, et al., 2013, p. 374-375). Kiltner et al. (2013) explains that the mechanisms of embodiment is determined by the mechanics of movement translated to the virtual avatar. Embodiment lets users inhabit a virtual avatar; however, immersion and presence allow for the mechanisms, a sense of being in a new world, possible. Presence is defined as a sense of “being there” or feeling that the environment you are in is real (Jennette et al., 2008). Jennette et al. (2008), argues that presence is related to embodiment and immersive environments, contributing to the overall sense of realness in a simulation. In gaming research, immersion refers to the quality of the simulation, while presence gives the user a sense of a “realness” (Jennett, et al., 2008; Slater, 2018). Although immersion and presence work in tandem in VR research, for this study, I refer to them as being mutually exclusive.

sburrell@uoregon.edu

2.4 En/De-Roling in VR

Character adaptation and relinquishing are not new concepts. From their roots in theatre and society, adapting and relinquishing roles is done for enjoyment or professional purposes (Gualeni et al., 2018). In a 2009 study, Stafford examined the behaviours of medical students when treating virtual patients. Her study showed that when these medical students treated virtual patients and their virtual patients died, students felt remorse, grief, and depression, a concept she coined as de-roling. Stafford defines the process of de-roling as allowing “the participants to discard parts of the role which may have been taken on, including inappropriate responsibility, and to restore them to a sense of who they are” (Stafford, 2009, p. 1084). Stafford’s (2009) definition of de-roling has challenged how we see effects from virtual characters from viewing virtual characters as a 2-D experience to now living and becoming the character themselves, however, more important, it has given an additional concept of character onboarding called en-roling (Gualeni, et al., 2018). Gualeni et al., (2018) examined the importance of de-roling from virtual characters by comparing debriefing (a concept in theater used to remove an actor from a character) and de-roling. Their initial findings came from the Live Action Role Playing (LARPing) game community and suggested a need for de-roling practices from virtual characters much like the debriefing in theater and LARPing. Although research on de-roling from virtual character is scarce in immersive media scholarship, studies have suggested en-roling affects an individual’s ability to change their attitudes and behaviors. En-roling is defined as becoming the character (Gualeni et al., 2020), although this was only a theoretical concept. En-roling suggest that for a while a user of VR takes on a virtual character and adapts or identifies to their new role as the user sees fit. The process of en-roling and de-roling become important when we examine how identification has an influence on VR effects and how these processes are established and negotiated.

3 QUALITATIVE STUDY

In this study, I exposed (N= 15) participants to a VR game called *Lone Echo* (Meta, 2017). This experience allowed participants move in zero gravity while exploring a space station and find who they were in a VR experience. The purpose of this step in my research was to allow for first-time VR users to explore and experience a highly immersive VR simulation and conveying their feelings and emotions while in the experience, generating concepts and themes to explore in future studies. Once the exposure of the VR experience was complete, participants were interviewed for about one hour and the data collected was recorded, transcribed, and analyzed. I used a thematic analysis to analyze the collected data and not surprisingly, I found major concepts and themes like, identification, embodiment, and immersion. However, surprisingly, a subcode from the data, “born again”, suggested some were actively participating in the VR simulation as someone they could create. Each participant who suggested this “born again” code expressed their wanting to navigate the VR simulation as a new person and expressed their wanting to explore a new world as someone who was not themselves. This then suggested that en-roling and de-roling were active during their identification process and still lingered in their interview. The analysis of the qualitative study provided a rich and useful perspective to how some first-time users of VR experienced a highly immersive simulation and how some wanted to create a new identity for themselves in that experience. Provided this data gives a glimpse into how VR effects may be established in experiences of first-time users through en-roling

and de-roling, this study also provides glimpse into these processes.

3.1 Discussion

The theories used in my research give a thorough understanding of how VR effects may happen when viewing how en-roling and de-roling have an influence on a user’s behavior or attitude. These theories have influenced my current study, examining how en-roling and de-roling my influence affect when exposed to a VR simulation. The qualitative study shows how further investigation on en-roling/de-roling can have a large impact on those who adapt and relinquish characters while playing as avatars in VR.

4 CONCLUSION

My research, although still on-going, provides some insight as to how and why users of VR may experience changes in attitudes and behaviors when taking on and leaving character roles. Currently, my research is in the pilot stages of establishing and sending out a survey to collect data from VR gamers and to examine their identification, embodiment, and presence in their gaming experiences as they take on character roles in VR. This survey study will examine the relationship of these variables and propel future experimental studies. However, my studies seek to answer these overall questions.

RQ1: How does en/de-roling processes negotiate VR effects?

RQ2: How do the processes of en/de-roling effect character identification?

RQ3: How does character identification effect VR effects when en/de-roling are not present?

REFERENCES

- Cohen, J., Appel, M., & Slater, M. D. (2016). Media, Identity, and the Self. In *Media Effect: Advances in Theory and Research* (4th ed., pp. 179–194).
- Gualeni, S., Vella, D., & Harrington, J. (2018). De-Roling from Experiences and Identities in Virtual Worlds. *Journal For Virtual Worlds Research*, 10(2), 1–19.
- Jennett, C., Cox, A. L., Cairns, P., Dhoparee, S., Epps, A., Tijs, T., & Walton, A. (2008). Measuring and defining the experience of immersion in games. *International Journal of Human Computer Studies*, 66(9), 641–661.
- Kilteni, K., Groten, R., & Slater, M. (2013). The Sense of embodiment in virtual reality. *Presence: Teleoperators and Virtual Environments*, 22(1), 373–387.
- Makransky, G., Terkildsen, T. S., & Mayer, R. E. (2019). Adding immersive virtual reality to a science lab simulation causes more presence but less learning. *Learning and Instruction*, 60(1), 225–236.
- Slater, M. (2018). Immersion and the illusion of presence in virtual reality. *British Journal of Psychology*, 109(3), 431–433.
- Stafford, F. (2005). The Significance of de-roling and debriefing in training medical students using simulation to train medical students. *Medical Education*, 39, 1083–1085.
- Nowak, K. L., & Fox, J. (2018). Avatars and computer-mediated communication: A review of the definitions, uses, and effects of digital representations. *Review of Communication Research*, 6, 30–53.
- Pimentel, D., & Vinkers, C. (2021). Copresence With Virtual Humans in Mixed Reality: The Impact of Contextual Responsiveness on Social Perceptions. *Frontiers in Robotics and AI*, 8(April), 1–11.